

Functional Testing

Validation of system behavior against requirements, business rules and observable expectations.

What is Functional Testing?

Functional Testing verifies whether a system feature behaves according to requirements, business rules, acceptance criteria and user needs. The focus is observable behavior: QA provides inputs, performs actions and compares the actual result with the expected result. It answers the question: “is the system doing what it is supposed to do?”. A login, for example, should not be validated only with correct credentials; we also need to observe invalid passwords, empty fields, incorrect formats, blocked accounts, messages, redirects and session persistence.

Black-box: behavior matters

Functional testing is often associated with Black Box Testing. Knowing the internal implementation is not mandatory to validate behavior: Input → System → Output. This does not mean code, APIs, databases and architecture are irrelevant; that knowledge helps investigate defects. In functional testing, however, the main objective remains verifying the behavior delivered by the product.

What do we want to validate?

Validate flows, business rules, mandatory fields, calculations, messages, permissions, navigation, persistence, integrations perceived by the user and different data combinations. Cover the happy path, alternative, negative and boundary scenarios, prioritizing what represents the greatest risk.

How to think like QA

Turn requirements into questions. In a login: what if the password is wrong? The email does not exist? Fields are empty? There are spaces before the email? The account is blocked? The user double-clicks Sign in? The session has expired? A core QA skill is moving from “I will test the button” to “I will test the behavior of the feature”.

Practical example — Shopping cart

Requirement: “the user can add products to the cart and change the quantity before completing the purchase”. Start by adding one product and validating name, quantity, price and subtotal. Then explore: add the same item again, change 1 to 3, zero or negative quantity, quantity above stock, remove the product, refresh the page and return later.

A simple technique to get started

Ask five questions: 1) What should work? 2) What should not be allowed? 3) What are the boundaries? 4) What happens if I interrupt the flow? 5) What happens next? For a field with a maximum of 100 characters, for example, try 99, 100 and 101.

Turning a scenario into a Test Case

CT001 — Log in with valid credentials

Precondition: registered and active user.

Test data: registered email and valid password.

Steps: 1) Access the login page. 2) Enter the email. 3) Enter the password. 4) Click Sign in. 5) Verify the page displayed.

Expected result: the user is authenticated and correctly redirected to the authenticated area.

Derived cases: CT002 — incorrect password; CT003 — nonexistent user; CT004 — empty fields; CT005 — invalid email; CT006 — blocked account.

When we find different behavior

If the expected behavior is to remain on the login page and display “invalid credentials”, but the system opens a blank page, we have Expected Result $\neq$ Actual Result. Investigate and, once the defect is confirmed, report a BUG with title, environment, precondition, steps, data, actual result, expected result and evidence.

Tools that can help

Jira, Xray and TestRail help with management; Postman supports API validation; DevTools helps with network and console analysis; SQL can validate persistence; Playwright, Selenium or Cypress can automate stable scenarios. Tools help execute tests, but they do not replace QA reasoning.

Quick checklist

Before finishing: main flow; alternative scenarios; invalid data; mandatory fields; boundaries; error and success messages; business rules; state after the action; interruptions; permissions; persistence; situations not explicitly described in the requirement.

QA Mindset

Do not test only what the system should do. Also test what the user may try to do. Good functional testing is not a long sequence of steps: it is an investigation guided by good questions about product behavior.